

Modeling and detection of false data injection attacks in cyber-physical distribution system with load aggregator interaction

Xinrui Liu ^{a,*}, Shubo Sun ^a, Yating Wang ^a, Zhiyuan Duan ^b, Xin Li ^a, Qiuye Sun ^a

^a College of Information Science and Engineering, Northeastern University, Shenyang, 110819, China

^b State Grid Shandong Electric Power Company Binzhou Power Supply Company, Binzhou, 256600, China

ARTICLE INFO

Keywords:

Load aggregator
Bidirectional interaction
False data injection attacks model
Attack detection
P-GAN

ABSTRACT

With the increasing number of users and multi-type loads accessing the cyber-physical distribution system (CPDS), the bidirectional interaction between the system becomes more and more frequent. The Load aggregator (LA) also plays an increasingly important role in the interaction process. However, the LA's high dependence on information and lack of security measures make it vulnerable to attacks, so this paper analyzes the interactive security of the LAs from two points. Considering the game process between the LAs and users from the attacker's point of view, this paper puts forward an attack strategy aiming cost function of the LAs, establishes a bi-level multi-objective programming attack model of false data injection attacks (FDIAs), and proposes an attack detection method based on the multi-state matching method and improved similar daily data preprocessing generative adversarial network (P-GAN) from the defender's point of view to defend against the above attack strategy. Furthermore, a hybrid detection mechanism combining event triggering and periodic detection is proposed to ensure response speed and adaptability of detection. The effectiveness of the proposed attack model and the detection method is verified by simulation analysis.

1. Introduction

The number of loads connected to the distribution power system continues to increase with the rapid development of the social economy, placing significant pressure on the system during peak load periods. Demand response (DR) has the advantages of power comfort and large-scale response, which may effectively relieve the shortage of system resources during peak hours by mobilizing resources such as load, energy storage, meanwhile enhancing the reliability and flexibility of the system [1].

As an emerging market participant in aggregating DR, the Load aggregator (LA) has the ability to evaluate and make full use of the potential of DR. It can also effectively integrate the scattered response resources to participate in system operation [2]. However, the integration of extensive demand-side resources, leads to a significant increase in high-dimensional uncertainty, posing challenges for the system's supervision of the LA [3]. From 2018 to 2023, the demand-side, including Portugal, France, South Africa, Japan and many other countries, suffered cyber attacks, among which the attack on Japanese photovoltaic power plants in 2023 was the most worrying, and the attackers have been able to control and interrupt the power grid operation through the cyber loopholes of demand-side equipment [4,5]. Due to the increasing integration of the cyber-physical distribution system

(CPDS) and the increasingly open power market environment, there will be more and more information interaction on the demand-side, which greatly increases the risk of attack on LAs and may cause worse impact than traditional cyber attacks [6]. Therefore, many scholars have conducted extensive research on CPDS demand-side attacks from different angles.

From the point of attacker's view, many loopholes on the demand-side can be exploited to achieve its malicious purpose. Such as the signal of the heating, ventilation, the DR message [7], the real-time price (RTP) [8]/[9–14], communication channel information [15], the signal of fast charging stations (FCS) [16]. Interestingly, it is very popular to manipulate or destroy the demand-side by injecting false data in RTP [9–11]. The attackers can manipulate the real-time price signal through false data injection attacks (FDIAs) in smart meters to increase power loss in the system [9]. The interaction model between real-time electricity price and total energy demand is formulated as a discrete nonlinear autonomous dynamic system in [10], and the influence of FDIAs on electricity price and voltages is also quantified. In [11], aiming at the uncertainty of FDIAs, a recursive least squares (RLS) method is proposed to evaluate the impact of FDIAs on electricity price. The FDIAs of the smart home management system is analyzed in [12], and the vulnerability of the demand-side that may be attacked

* Corresponding author.

E-mail address: liuxinrui@ise.neu.edu.cn (X. Liu).

<https://doi.org/10.1016/j.segan.2024.101533>

Received 13 May 2024; Received in revised form 20 August 2024; Accepted 2 October 2024

Available online 5 October 2024

2352-4677/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Comparison between this paper and other studies.

Ref	Both attack and detection model	Attack model				Detection model	
		Position		Target	Method	Target	Method
		DS	LA				
This paper	✓	✓	✓	LA cost function	FDI	Price and load	Machine learning
Yang [8] 2024	×	✓	×	RTP	FDI	–	–
Raman [7] 2019	✓	✓	✓	DR message	Multi-type	–	–
Giglou [9] 2021	×	✓	×	RTP	FDI	–	–
Jhala [10] 2022	×	✓	×	RTP	FDI	–	–
Zhang [11] 2024	×	✓	×	RTP	FDI	–	–
Sajeev [12] 2020	×	✓	✓	RTP	FDI	–	–
Liu [13] 2018	✓	✓	×	RTP	Coordinated	Price and load	POMDP
Zhou [14] 2019	✓	✓	×	RTP	Coordinated	Price and load	POMDP
Elrawy [15] 2023	✓	✓	×	Channel information	MITM	Channel information	Machine learning
Warraich [16] 2023	✓	✓	×	The signal of FCS	DDOS	Power demand	Machine learning

under the aggregator is evaluated. However, the attack model for the interactive process is not given. However, the above references mainly focus on modeling the influence of FDIAs on the CPDS, ignoring the attack modeling of the interactive process in the system. According to the above analysis, as a part of the demand-side with high interactivity and low security, LA is necessary to model its interaction process when considering the impact of attacks on the demand-side. However, there are few models of this process in the current references. Considering attackers may exploit this vulnerability to attack CPDS, the interaction between users and the LA is studied in this paper. And a bi-level multi-objective programming model of FDIAs considering the LA operation is established, which may cause an abnormal response on the demand-side, resulting in line overload and the LA economic loss.

From the defenders' point of view, many different demand-side attack detection methods have been proposed. To solve the cooperative network attack problem, a continuous state detection framework based on a partially observable Markov decision process (POMDP) is proposed in [13]. In [14], a long-term detector based on continuous state, partially observable Markov decision process is constructed and a short-term detector based on Binary Logistic Regression is added to detect cooperative attacks. A Hybrid Network Intrusion Detection System based on machine learning and signature is proposed in [15] for detecting man-in-the middle (MITM) attacks. A machine learning method is proposed for the early detection of micro-grid energy management system with electric vehicles in [16], so as to prevent the economic losses of users and utilities caused by distributed denial of service attacks (DDOS). However, the detection methods in [13–16] do not consider the scenario that false data is injected to attack the load aggregator. In addition, the machine learning detection methods in [15,16] do not consider the diversity and seasonality of load, and no preprocessing operation is given. In order to solve the above problems, a preprocessing generative adversarial network (P-GAN) detection method is proposed for the first time in this paper (see Table 1).

(a) From the attacker's point of view, an attack strategy against the cost function of the LA is proposed, and a bi-level programming model including the Stackelberg game process between the LAs and users is established. The implementation of the proposed attack strategy is hidden, and the established attack model not only makes the demand response target in the LA contract unable to be successfully completed but also may unbalance the source and load, thus destroying the system.

(b) From the defenders' point of view, an improved LA detection method based on P-GAN is proposed. This method can detect the attack through the change of load and electricity price, and weaken the adverse impact of the attack by issuing the revised electricity price. This method uses multi-state matching and improved similar day method to process data, thus significantly improving the performance of machine learning. The proposed P-GAN detection method does not need to define the threshold, thus improving the detection accuracy.

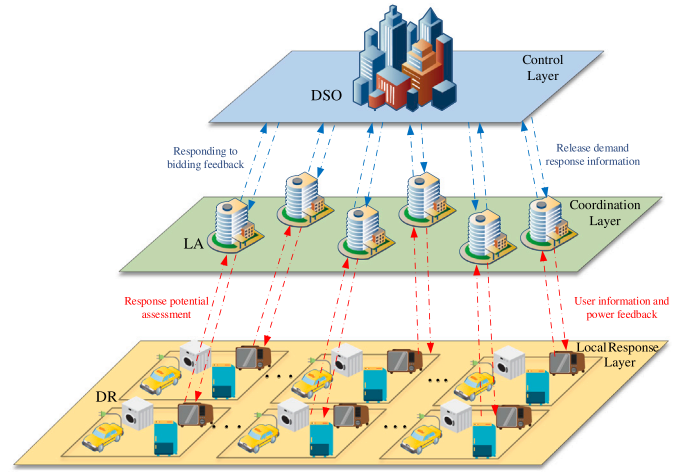


Fig. 1. The operational mode in LA.

(c) Considering that the traditional periodic detection mechanism is not enough to deal with sudden attacks effectively, a hybrid detection mechanism combining event triggering and periodic detection is proposed. The above mechanism not only ensures continuous detection, but also improves the efficiency and adaptability of the detection process.

2. The bidirectional interaction process of CPDS

The real-time adjustment of the distribution system operator (DSO) does not involve a large number of small and medium-sized users of CPDS. The LA can instruct these users to participate in the price response [17], thus optimizing the utilization of demand-side resources. The users adjust their power consumption behaviors according to price changes, thereby reducing the scheduling pressure on the system.

2.1. The LA interactive process model

The LA [17] acts as an intermediary between the DSO and DR, providing load balancing and other basic services for CPDS on behalf of users. The LA operation framework consists of DSO, LAs, and users, as illustrated in Fig. 1.

The user's load demand Q in the future period is guided within the peak hour threshold Q_{thsd}^+ and the valley hour threshold Q_{thsd}^- by adjusting the electricity price through the LA [18]. The relationship can be expressed as:

$$Q_{thsd}^- \leq Q \leq Q_{thsd}^+ \quad (1)$$

The DR resources of the same node are controlled by the same LA by default in this paper. And the LA only interacts with consumers who have signed contracts through the communications infrastructure. The advanced metrology infrastructure (AMI) realizes bidirectional information flow between the LA and its participating users, serving as an effective communication medium. If the active user is not the contract load of the LA or any other LA, in principle, the DR will not affect the electricity price of the active user who has signed the contract.

As a profit-driven entity in the real-time interactive DR market, the LA can dynamically adjust the electricity price during different periods according to the total load usage of users in a specific period, and ensure that its behavior will not destroy the stability of the system or exceed the maximum load limits. Therefore, the users' load response is guided to ensure that the maximum revenue is obtained from the LA. The revenue can be expressed by formula (2)–(3):

$$F_{i,t} = Q_{i,t}p_{i,t} - Q_{i,t}\sigma_{i,t} - C_{i,t}(Q_{i,t}) \quad (2)$$

$$C_{i,t}(Q_{i,t}) = a_{i,t}(Q_{i,t})^2 + b_{i,t}Q_{i,t} + c_{i,t} \quad (3)$$

where $F_{i,t}$ is the revenue of i th LA in period t . The $Q_{i,t}$ actually purchased by i th LA in period t is equal to the aggregate of the actual response from its subordinate users, namely, $Q_{i,t} = \sum_{n=1}^{N_{i,t}} \Delta Q_{i,n,t}$. $N_{i,t}$ is the number of participating users of i th LA in period t .

$Q_{i,t}p_{i,t}$ means the revenue generated by the real-time sale of electricity to users by LA, $p_{i,t}$ is the selling price adjusted by LA dynamically according to the period response. $Q_{i,t}\sigma_{i,t}$ is the actual expenditure on procuring power in the electricity market, $\sigma_{i,t}$ is the actual purchasing price of power by the LA. $\sigma'_{i,t}$ is the original electricity price purchased by LA from the operator, $\Delta\sigma_{i,t}$ is The compensation price given to LA by the operator according to the load response. When the load response reaches the operator's demand, the operator will give LA a reduction based on $\Delta\sigma_{i,t}$, as $\sigma_{i,t} = \sigma'_{i,t} - \Delta\sigma_{i,t}$. Otherwise, LA will be purchased at the original price, as $\sigma_{i,t} = \sigma'_{i,t}$. $C_{i,t}(Q_{i,t})$ is the operation and maintenance costs of the LA [19]. $a_{i,t}$, $b_{i,t}$ and $c_{i,t}$ are the cost factors, which depend on the operation and maintenance activities, thereby show fluctuations in time.

2.2. Modeling of user behavior analysis

The LA will modify the user's electricity consumption habits through price guidance, and ensure that the normal needs of users will not be affected. However, the effect of user's satisfaction should also be considered when the user responds to the electricity price. The user's dissatisfaction function $\varphi(\Delta Q_{i,n,t}, \rho_{i,n})$ is defined to reflect the user's satisfaction with this response when it is higher or lower than the ideal response:

$$\varphi(\Delta Q_{i,n,t}, \rho_{i,n}) = \rho_{i,n} |\Delta Q_{i,n,t} - Q_{i,n,ideal}| \quad (4)$$

where $\rho_{i,n}$ is the dissatisfaction coefficient of the n th user belong to the i th LA, which consists of two parts: one part is a consumer attitude coefficient, which reflects customers' attitude towards load interruption, and the other part is a load type-related factor, which indicates different types of customers' willingness to respond. $\Delta Q_{i,n,t}$ is the excitation response of the n th user during period t , $Q_{i,n,ideal}$ is the ideal response of the n th user.

After the price of other periods is published by the LA, users will make the optimal response strategies to minimize their electricity expenditure. Users will also lose some satisfaction with power consumption while adapting to the electricity price. Therefore, the user's cost is:

$$U_{i,n,t} = p_{i,t}\Delta Q_{i,n,t} + \varphi(\Delta Q_{i,n,t}, \rho_{i,n}) \quad (5)$$

3. Malicious cyber attack model for LA

Fig. 2 illustrates the two malicious intrusion processes of the attacker. One is that the attacker intercepts the cost function information through the LA communication vulnerability and tampers with the cost function parameters; The other is that the attacker invades LA's system through the device system vulnerability of LA, releases malicious software to find the cost function and injects false parameters to tamper with it. It should be assumed that the attacker has advanced knowledge of the weaknesses of the target system in this paper, including general configuration and default defense mechanisms (e.g., built-in firewalls, rate limits, traffic policing) [20]. The cost function and revenue function of the attacker are shown in the following sections.

3.1. The cost function of the attacker

The attacker manipulate data through vulnerabilities in the communication system, and the cost of attack is positively correlation to the difficulty of exploiting these vulnerabilities. The score index of *exploiting complexity* in the common vulnerability scoring system (CVSS) [21] is utilized in this paper to quantify the level of vulnerability difficulty. The higher the score, the more difficult it is to exploit the vulnerability, and the greater the cost of implementing the attack.

This paper also refers to the vulnerability parameters of communication equipment in the industrial internet security response database [22]. The cost of the attacker exploiting the vulnerability to invade and manipulate the i th LA can be formulated as:

$$C_i^a = \frac{g_i}{\sum_{i=1}^M g_i} \quad (6)$$

where g_i represents the data utilization complexity score of the i th LA. M is the number of LAs.

3.2. The revenue function of the attacker

(1) The system's loss function

The system will encounter power flow fluctuations and subsequent losses due to FDIAs. The active power flow volatility λ_t of the total line in period t is selected as the attacker's revenue function.

$$\lambda_t = \sum_{l=1}^L \lambda_{l,t} \quad (7)$$

$$\lambda_{l,t} = \frac{\sum_{i=1}^M \Delta p_{l,t}^i}{p_{l,b}} \quad (8)$$

where $\lambda_{l,t}$ is the fluctuation of active power flow in the line l . $p_{l,b}$ is the active power flow of the line l under normal operating conditions of the system. $\Delta p_{l,t}^i$ represents the active power flow difference induced on the line due to load adjustment of the i th LA, which is given as

$$\Delta p_{l,t}^i = \sum_{i=1}^k G_{l-i} \times \Delta Q_{i,t}^{act} \quad (9)$$

$$G_{l-i} = \frac{1}{x_l} M_l^T X_i = \frac{X_{mi} - X_{ni}}{x_l} \quad (10)$$

where $\Delta Q_{i,t}^{act}$ is the load response increment of the attacked i th node in period t . G_{l-i} is the variation in active power flow of line l resulting from a unit change in active power at the i th node.

(2) The LA loss function

If the LA fails to deliver the agreed response load to the operator within the specified period t as stipulated in the contract, the operator will seek compensation for the breach of contract from LA. Moreover, the purchase price of the service will still be paid to the LA at the actual price. The attack will cause economic losses of the LA, and the attacker

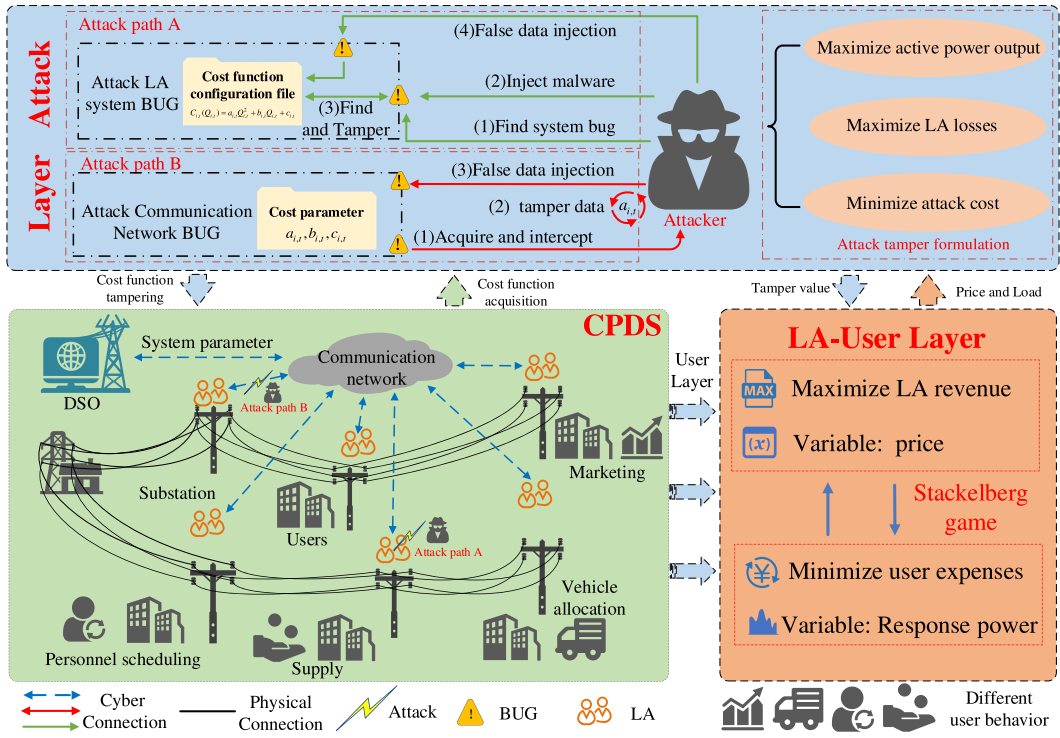


Fig. 2. The flow chart for the attack.

will profit from it. The economic loss function R_t of the LA can be formulated as:

$$R_t = \sum_{i=1}^{M_t} (p_{pen} Q_{i,s} + (p_{i,t} - \sigma_{i,t}) \Delta Q_{i,t}^{act}) \quad (11)$$

where M_t is the number of LAs attacked during the period. p_{pen} is the price of compensation cost for breach of contract. $Q_{i,s}$ is the difference between the actual load increment of the i th node and the threshold value.

$$Q_{i,s} = \begin{cases} |\Delta Q_{i,t}^{act}| - Q_{i,thsd}^+, & |\Delta Q_{i,t}^{act}| > Q_{i,thsd}^+ \\ 0, & |\Delta Q_{i,t}^{act}| \leq Q_{i,thsd}^+ \end{cases} \quad (12)$$

$$p_{pen} = K(1 + \frac{Q_{i,s}}{Q_{i,t}})p_0 \quad (13)$$

where K is the linear coefficient, p_0 is the compensation price.

3.3. The bi-level programming attack model

3.3.1. Attack model

The constructed bi-level attack model is illustrated in Fig. 3. The upper attacker manipulates the cost function parameters of the LA in the information flow transmitted by the DSO, leading to inaccurate scheduling of the system. The pricing model for the LA and the real-time response model for various types of users are established in the lower level. The decision variables of the upper level are regarded as fixed parameters in the lower-level model, and the revenue of the LA is affected by users' response-ability.

(1) Attack programming level

The parameters of the LA cost function of each period are influenced by many uncertain factors, such as the operation and maintenance of metering equipment, data transmission, market feedback and so on. The attackers may inject attack tamper value $\Delta a_{i,t}$ into the information flow transmitted by operators to each target LA. The change of the measured value caused by the change of parameters is expressed as

$$z_a = z + \Delta \hat{a}_{i,t} \quad (14)$$

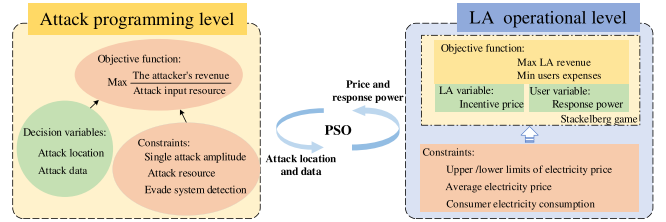


Fig. 3. Bi-level model.

where $\Delta \hat{a}_{i,t}$ is the correlation between the parameter and attack vectors, denoted as $\Delta \hat{a}_{i,t} = f(V, \theta, p_{i,t}) \Delta a_{i,t}$. f is the correlation function. V and θ are the voltage amplitude and voltage phase angle, respectively. The anticipated alteration of state variables caused by FDIA is:

$$x_a = x + \varpi_{i,t} \quad (15)$$

where $\varpi_{i,t}$ is the error vector of the state variable caused by FDIA. The residual following the attack can be expressed as:

$$\|r\| = \|z - Hx + \Delta \hat{a}_{i,t} - H\varpi_{i,t}\| \quad (16)$$

Assuming that $\Delta \hat{a}_{i,t} = H\varpi_{i,t}$ in (16), then $\|r\| < \tau$ is feasible, and the constructed FDIA can cheat the system detection. τ is the threshold for largest normalized residual check. The objective function of the upper attacker is defined as the ratio W of attack revenue to attack resources, aiming to maximize the effectiveness of attacks with limited resources:

$$\max W = [\omega \lambda + (1 - \omega)N(R)] / \sum_{i=1}^{M_a} C_i^a, \forall i \in B \quad (17)$$

where $N(\cdot)$ is a normalized function. ω is the specific gravity coefficient. M_a is the number of LAs attacked. The constraint of attack resources is:

$$\alpha_{B,i} \cdot a_{i,b} \leq \Delta a_{i,t} \leq \beta_{B,i} \cdot a_{i,b}, i \in B \quad (18)$$

$$\sum_{i \in B} (C_{B,i} \cdot \delta_{B,i}) \leq RE \quad (19)$$

$$\sum_T |\Delta a_{i,t}| \leq \nu a_{i,b} \quad (20)$$

where B is the set of attacked LAs. $a_{i,b}$ is the parameter of cost function, which fluctuates by $\pm 50\%$ in each period. The attacker estimates $\alpha_{B,i}$ and $\beta_{B,i}$ based on empirical knowledge. ν is the constraint coefficient. T is the set of attacked periods. $C_{B,i}$ is the cost of the attack on the i th LA. $\delta_{B,i}$ is a binary variable whose value is 1, indicating that the LA is attacked. RE is the maximum attack resources.

(2) The LA operational level

The lower LA mitigates load fluctuations according to the inaccurate service price, calculates the compensation electricity price to maximize its revenue, and obtains responsive power corresponding to users' consumption behaviors. The optimal model for the i th LA is:

$$p_{i,t}^* = \argmax_{p_{i,t}} F'_{i,t} = \Delta Q_{i,t}^{act} p_{i,t} - \Delta Q_{i,t}^{act} \sigma_{i,t} - C_{i,t} \quad (21)$$

$$C_{i,t} = a_{i,t} \left(\Delta Q_{i,t}^{act} \right)^2 + b_{i,t} \Delta Q_{i,t}^{act} + c_{i,t} \quad (22)$$

$$p_{i,t}^l \leq p_{i,t} \leq p_{i,t}^u \quad (23)$$

$$\sigma_{i,t} \geq 0 \quad (24)$$

For N_i users belonging to the i th LA:

$$\Delta Q_{i,t}^{act*} = \sum_{n=1}^{N_i} \argmin_{U_{i,n,t}} U_{i,n,t} \quad (25)$$

$$\sum_{t=1}^{24} \Delta Q_{i,n,t} = Q_{i,n} \quad (26)$$

$$0 \leq \Delta Q_{i,n,t} \leq \Delta Q_{i,n}^u \quad (27)$$

where $Q_{i,n}$ is the total translatable load of the n th user within a day. $\Delta Q_{i,n}^u$ is the maximum threshold for the translatable load response variation within a single period.

3.3.2. The solution for attack model

The LA level model in (21) is a stackelberg game model, which can be transformed into an integer programming problem to improve operational efficiency.

The optimization problem can be eliminated and a new constraint relationship between $\{\Delta Q_{i,n,t}\}$ and $\{\sigma_{i,t}\}$ can be obtained by replacing the linear programming formula (25)–(27) with its Karush–Kuhn–Tucker (KKT) condition, which are given as follows:

$$p_{i,t} - \mu_{i,n,t}^{\max} - \mu_{i,n,t}^{\min} - \eta_{i,n} = 0 \quad (28)$$

$$0 \leq \mu_{i,n,t}^{\min} \perp \Delta Q_{i,n,t} \geq 0 \quad (29)$$

$$0 \leq \mu_{i,n,t}^{\max} \perp \Delta Q_{i,n,t} - \Delta Q_{i,n}^u \leq 0 \quad (30)$$

where $\mu_{i,n,t}^{\max}$, $\mu_{i,n,t}^{\min}$ is the dual variable of (27). The constraints (29) and (30) are complementary relaxation conditions, denoted by the symbol $\perp$, indicating that the product of two expressions is equal to 0. $\eta_{i,n}$ is the dual variable of (26). The method of [23] is used to introduce Boolean variables $\theta_{i,n,t}^{\max}$ and $\theta_{i,n,t}^{\min}$, (29)–(30) can be converted into linear inequalities:

$$0 \leq \mu_{i,n,t}^{\min} \leq A \theta_{i,n,t}^{\min} \quad (31)$$

$$0 \leq \Delta Q_{i,n,t} \leq A(1 - \theta_{i,n,t}^{\min}) \quad (32)$$

$$0 \leq \Delta Q_{i,n}^u - \Delta Q_{i,n,t} \leq A \theta_{i,n,t}^{\max} \quad (33)$$

$$A(\theta_{i,n,t}^{\max} - 1) \leq \mu_{i,n,t}^{\max} \leq 0 \quad (34)$$

where A is an infinite positive number. According to the duality theorem, the objective function (25) is transformed into:

$$U_i = \sum_n (\eta_{i,n} Q_{i,n} + \sum_t \mu_{i,n,t}^{\max} \Delta Q_{i,n}^u) \quad (35)$$

The objective function (21) is equivalent to under the conditions of KKT:

$$\max \left(\sum_n \eta_{i,n} Q_{i,n} + \sum_n \sum_t \mu_{i,n,t}^{\max} \Delta Q_{i,n}^u - \sum_t \left(\Delta Q_{i,t}^{act} \sigma_{i,t} + C_{i,t} \right) \right) \quad (36)$$

The established bi-level programming problem can be solved by using the particle swarm optimization algorithm and YALMIP+CPLEX, then the optimal attack scheme can be obtained.

4. Malicious data defense mechanism

The above attack model can induce abnormal load responses by manipulating the parameters of the LA cost function, which will lead to overload and even operation interruption. The long-term effect of the attack is harmful, and traditional bad data detection cannot detect it in time. In this section, a load feature time sequence detection method according to the attack model is proposed.

The load data in a period of time is regarded as a matrix showing the correlation between time and space. The spatial correlation of data is expressed by Kirchhoff's law and system topology, while the temporal correlation of data is determined by factors such as the power consumption behaviors of users and the momentum of the system in continuous time. The consideration of the attack data often leads to prioritizing spatial correlation exclusively, ignoring temporal correlation and system dynamics. The paper thus employs time-correlated detection technology to construct a defensive framework for identifying malicious data.

The basic concept of this method is to utilize the regional load, electricity price, and their influencing factors on a typical day as the reference point. Subsequently, the P-GAN is trained on preprocessed data sets, which spans a period of time, such as three years. Then, the price of LA will be checked and adjusted in the future (for example, on a certain day). The core of this method is the sample preprocessing process, which effectively utilizes the seasonal variation of regional power consumption, thus reducing detection time and improving the accuracy. This paper proposes the P-GAN for detection, which incorporates multi-state matching and improved similar day extraction techniques to effectively identify and correct abnormal electricity price resulting from FDIAs.

4.1. The method of multi-state matching

4.1.1. The model of time windows

Considering the intrinsic relationship among the state variables of each node within the system, the historical state constructed by the CPDS is organized chronologically to form a multidimensional time series matrix $\mathbf{T}_{q \times p}$ with q rows and p columns. q is the number of states that the node contains. The number of nodes with AMI is p . The system states refer to the amplitude and phase angle of the node voltage. The number of collected states is determined according to the interval of dynamic electricity price, which is illustrated at 0.5 h in this paper.

The set of L groups of continuous state collection points in $\mathbf{T}_{q \times p}$ is used as the state model of CPDS. L represents the length of the time window. The sliding time window modeling based on the state model is illustrated in Fig. 4. The state x are all $q \times 1$ dimension matrices. In the historical database, $(\mathbf{X}_v)_{q \times L}$ is assumed the current state matrix.

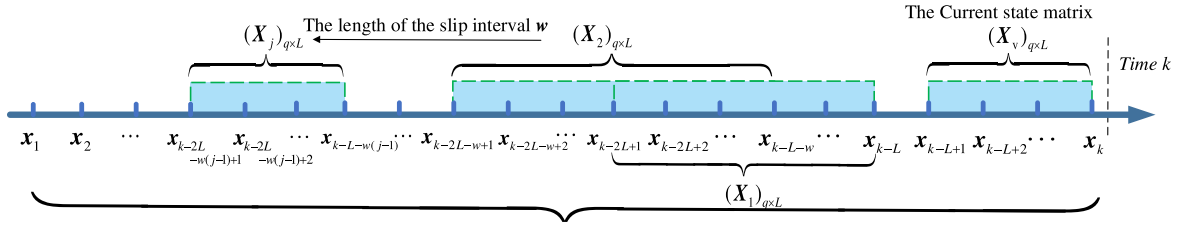


Fig. 4. Model of sliding time window.

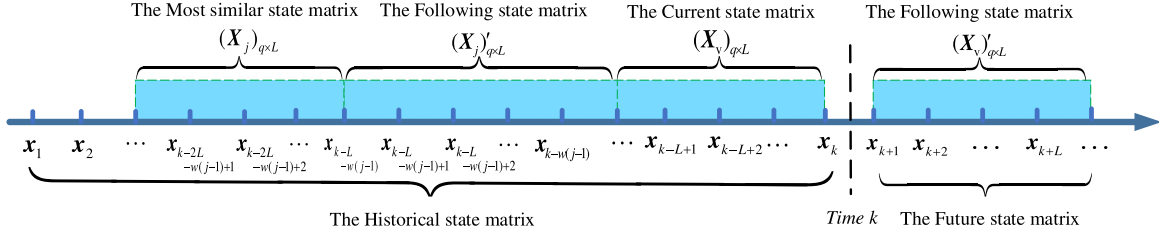


Fig. 5. Multi-state matching method.

The matrix adjacent to the current state matrix is denoted as $(X_1)_{q \times L}$. The state matrix under the j th time window can be obtained by sliding $(X_1)_{q \times L}$ backward along the time axis.

$$X_j = [x_{k-2L-w(j-1)+1}, x_{k-2L-w(j-1)+2}, \dots, x_{k-L-w(j-1)}] \quad (37)$$

where $j \in (1, \lfloor (P-2L)/w+1 \rfloor)$. The integer $\lfloor (P-2L)/w+1 \rfloor$ is the floor of $[(P-2L)/w+1]$. w is the length of the sliding interval.

The discrete time series can be divided into a set of multiple $q \times L$ dimension historical state matrices through the model of sliding time window. i.e. $X = [X_1, X_2, \dots, X_j, \dots, X_{\lfloor (P-2L)/w+1 \rfloor}]$. The state matrix $(X_j)_{q \times L}$ with the highest similarity to the current state matrix $(X_v)_{q \times L}$ is then determined from X . The method is shown in Fig. 5.

4.1.2. The process of state matrix matching

The initial matching of the state matrix is completed through similarity analysis, followed by exact matching achieved through cluster analysis. Any two node state matrices $A' = [a']_{q \times L}$ and $B' = [b']_{q \times L}$ are selected from the historical database. Based on the Pearson correlation coefficient, the similarity measurement index is as follows:

$$\text{sim}(A', B') = \frac{1}{q} \sum_{i=1}^q \left| \frac{\sum_{j=1}^L (a'_{ij} - \bar{a}_i)(b'_{ij} - \bar{b}_i)}{\sqrt{\sum_{j=1}^L (a'_{ij} - \bar{a}_i)^2} \times \sqrt{\sum_{j=1}^L (b'_{ij} - \bar{b}_i)^2}} \right| \quad (38)$$

where $\bar{a}_i = \frac{1}{L} \sum_{j=1}^L a'_{ij}$, $\bar{b}_i = \frac{1}{L} \sum_{j=1}^L b'_{ij}$.

When the given similarity measurement threshold $\varphi \in (0, 1)$ and $\text{sim}(A', B') \geq \varphi$, The node state matrices A' and B' are mutually coupled similar matrices. The greater the value of $\text{sim}(A', B')$, the higher the average similarity among the states of each node in the state matrix. The optimal parameters w and L are selected based on the following conditions [24]:

$$\text{num}\{\text{sim}(X_j^{\bar{w}, \bar{L}}, X_v) \geq \varphi\} \geq \text{num}\{\text{sim}(X_j^{w, L}, X_v) \geq \varphi\} \quad (39)$$

where $X_j^{\bar{w}, \bar{L}}$ represents the set of state matrices that satisfy the conditions with the optimal parameters. $X_j^{w, L}$ represents the set of state matrices that satisfy the condition under any parameter. num represents the number of elements in each set, i.e., the number of state matrices.

The Density-Based Spatial Clustering of Applications with Noise algorithm (DBSCAN) [25] is applied to the state matrix set obtained from the similarity analysis to achieve the optimal matching result of the state matrix. The Mahalanobis distance was employed as the clustering metric, and K clusters were derived based on the results of DBSCAN. The minimum sum of Mahalanobis distances between the center point

of each cluster and other state matrices within the cluster is guaranteed if the conditions (40) are satisfied by the $C = \{C_1, C_2, \dots, C_K\}$.

$$M(C_K) = \min \sum_{i=1}^{R_K} D(C_K, U_i) \quad (40)$$

where R_K represents the number of state matrix samples in the cluster of class K . U_i is the state matrix. $M(C_K)$ is calculated as the sum of Mahalanobis distances at the centroid of the class K cluster.

The current state matrix $(X_v)_{q \times L}$ is compared with other state matrices, and the degree of difference is sorted. The additional information of each node in the corresponding period is acquired from the matrix with the minimum difference.

4.2. Improved similar day selection method

In part A, the internal relationship between the states of CPDS nodes is considered, and the regional data of each LA in the corresponding period of the matching matrix are obtained. However, the daily load is greatly influenced by meteorological and date-related factors [26]. Therefore, similar periods should be further screened according to the meteorological, dates, and other factors specific to each LA region.

The date distance factor is introduced in this paper based on the shape similarity [27]. Meanwhile, the load is categorized into five groups [28] and weighted based on their respective proportions. Finally, combining meteorological factors, date factors, and load proportion, an improved weighted grey correlation method for similar day selection is proposed.

Step1 (Construct the feature matrix): Factors affecting daily loads such as temperature, humidity, wind speed, precipitation and date type are selected to form daily feature vector:

$$E_i = [e_{i1}, e_{i2}, \dots, e_{in}], i = 0, 1, \dots, m \quad (41)$$

where m is the total number of historical days. e_{in} is the n th influencing factor of the i th historical sample day.

Step2 (Dimensionless processing): Referring to E_0 as the basic value, the dimensionless feature vector $E'_i = [\frac{e_{i1}}{e_{01}}, \frac{e_{i2}}{e_{02}}, \dots, \frac{e_{in}}{e_{0n}}]$ is derived.

Step3 (Difference sequence solution): The reference sequence E'_0 and the comparison sequence E'_i are utilized to solve the difference sequence, resulting in:

$$\Delta E_i(k) = |E'_0(k) - E'_i(k)|, k = 1, 2, \dots, n \quad (42)$$

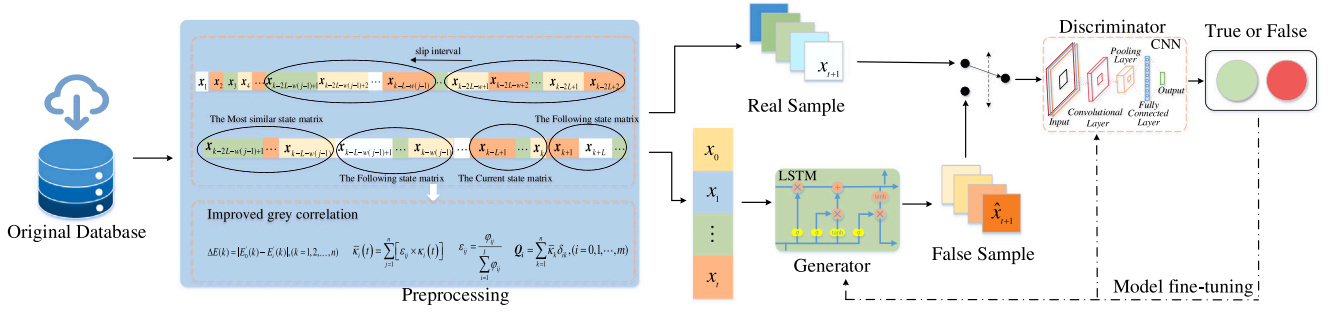


Fig. 6. P-GAN detection model.

Step4: Construct the correlation coefficient matrix and calculate $\delta_{ik} = \frac{\Delta E_{\min}(k) + \rho \Delta E_{\max}(k)}{\Delta E_i(k) + \rho \Delta E_{\max}(k)}$. $\Delta E_{\max}(k)$ and $\Delta E_{\min}(k)$ are the largest and smallest value of $\Delta E_i(k)$ respectively, we have

$$\delta = \begin{bmatrix} \delta_{01} & \delta_{02} & \cdots & \delta_{0n} \\ \vdots & \vdots & \vdots & \vdots \\ \delta_{m1} & \delta_{m2} & \cdots & \delta_{mn} \end{bmatrix} \quad (43)$$

The resolution coefficient, denoted as ρ is typically set to 0.5.

Step5: The weight of each influencing factor is adjusted according to the proportion of the five types of loads.

$$\bar{\kappa}_i(t) = \sum_{j=1}^n [\varepsilon_{ij} \times \kappa_i(t)] \quad (44)$$

$$\kappa_i(t) = \frac{L_i}{L_a + L_b + L_c + L_d + L_e}, i = a, b, c, d, e \quad (45)$$

where ε_{ij} is the weight of the j th influencing factor affected by the i th class load. $\kappa_i(t)$ is the proportion of five different types of load. n is the number of influencing factors. L_a is the load of urban residents. L_b is the urban commercial load. L_c is the industrial load. L_d is the traffic load, L_e is other types of load.

The load data was analyzed to determine the correlation and degree of influence of various factors [29]. The correlation coefficient ε_{ij} is expressed by the following formula.

$$\varepsilon_{ij} = \frac{\varphi_{ij}}{\sum_{i=1}^I \varphi_{ij}} \quad (46)$$

$$\varphi_{ij} = \frac{\text{cov}(P_i, Y_j)}{\sqrt{D(P_i)} \sqrt{D(Y_j)}} \quad (47)$$

where I is the total number of load categories. $\text{cov}(\cdot)$ is the covariance function. $D(\cdot)$ is the mean square error function. P_i is the historical daily load sequence of i th type loads, and Y_j is the historical daily mean sequence of the j th influencing factor. The weight vector of each influencing factor is as follows:

$$K = [\bar{\kappa}_1(t) \quad \bar{\kappa}_2(t) \quad \cdots \quad \bar{\kappa}_n(t)] \quad (48)$$

The weighted grey correlation coefficient matrix is then derived:

$$\delta' = \begin{bmatrix} \bar{\kappa}_1(t) \delta_{01} & \bar{\kappa}_2(t) \delta_{02} & \cdots & \bar{\kappa}_n(t) \delta_{0n} \\ \vdots & \vdots & \vdots & \vdots \\ \bar{\kappa}_1(t) \delta_{m1} & \bar{\kappa}_2(t) \delta_{m2} & \cdots & \bar{\kappa}_n(t) \delta_{mn} \end{bmatrix} \quad (49)$$

Step6 (Calculate the grey-weighted correlation degree): The weighted grey correlation degree can be obtained by summing the row vectors of matrix δ' respectively:

$$Q_i = \sum_{k=1}^n \bar{\kappa}_k \delta_{ik}, i = 0, 1, \dots, m \quad (50)$$

The date distance factor a^k is introduced here. a is the attenuation coefficient, $k = \{k_1, k_2, \dots, k_m\}$ indicates the difference between the historical day and the predicted day. The improved weighted grey correlation degree based on the date factor can be obtained:

$$Q' = \begin{bmatrix} Q_0 \\ Q'_1 \\ \vdots \\ Q'_m \end{bmatrix} = \begin{bmatrix} Q_0 \\ a^{k_1} Q_1 \\ \vdots \\ a^{k_m} Q_m \end{bmatrix} \quad (51)$$

Step7: The improved weighted grey correlation degree of each historical day is sorted. Then, m' samples with high correlation degree and n days' data (10 days selected for this paper) before and after the samples are selected to form the sample set used in the next section.

The improved weighted grey correlation analysis method can adjust the weight of various influencing factors according to different load ratios, and fully consider the time and distance factors, which significantly improve the effectiveness of sample extraction and recombination of similar days.

4.3. The FDIAs detection method based on P-GAN

The data component in the load vector is regarded as the feature value of the normal or attacked load, and the FDIAs can be identified by classification detection. The required historical data samples tested in this paper have obvious seasonal characteristics. Traditional GAN not only takes a long time to process complex sample data for different seasons and holidays but also cannot achieve ideal detection results, while P-GAN can complete the detection task well. Meanwhile, compared with traditional predictive detection methods, the P-GAN method eliminates the need for manual selection of an optimal threshold, effectively mitigating the detection method's reliance on threshold values. The detection model is illustrated in Fig. 6.

According to the analysis in Sections 2 and 3, when the LA is attacked at time k , it will set the false electricity price through the Stackelberg game model. Meanwhile, P-GAN is used to determine whether the system is attacked by identifying whether the dynamic electricity price released by the LA deviates from the standard obviously.

Considering the existence of multiple time series data, there will be different types of features, and the target data will depend on these. Therefore, the generator (G) in this paper adopts the multi-input single-output LSTM network to track the previous data in a certain period and output the predicted target data, and the discriminator (D) will use the CNN network.

The input of the generator consists of the influencing factors of the t period, it can be represented by $X = \{x_1, \dots, x_t\}$. The vector x_k in X consists of 9 features, as device measurements (typically from AMI), the LA energy storage device information, real-time electricity price market conditions, user load information, electricity price, and environmental factors (dry bulb temperature, wet bulb temperature and humidity), which can be expressed as $x_k = [x_{k,i}]_{i=1}^9, k = 1, \dots, t$. By training the generator and discriminator, the pre-processed data can be input into G as the sample data to obtain the predicted data (i.e. $\hat{x}_{t+1.5}$) after time t . The historical load/electricity price is taken as an input to D , which determines that the system is not under attack when the condition is

met by comparing the predicted data with the actual data (i.e. $x_{t+1,5}$), and if the condition is not met, the D determines that the system is under attack. The objective function of the generative adversarial network in this paper is:

$$\min_G \max_D V(D, G) = E_{k \sim p(p_{i,t})} [\log D(p_{i,t})] + E_{k \sim p(x_k)} [\log D(G(x_k))] \quad (52)$$

Algorithm 1 P-GAN training process

- Input: Maximum Iterations N ,
Historical status data $X = \{x_1, \dots, x_m\}$
Output: Predicted the LA response electricity price $\hat{x}_{t+1,5}$;
1. Perform multi-state matching on X ;
 2. Obtain the LA data from the matching matrix with the corresponding period;
 3. The improved weighted grey correlation analysis method was used to further filter the new X ;
 4. Initialize the generator θ_g and discriminator θ_d ;
 5. **for** number of training iterations **do**
 6. **for** k step **do**
 7. Small batch samples of X as input to generator G ;
 8. X_{real} and X_{fake} are input to discriminator D ;
 9. Calculate the cost function of the discriminator;
 10. The discriminator parameters θ_d were updated by the Adam gradient descent algorithm;
 11. **end for**
 12. Small batch samples of X as input to generator G ;
 13. Calculate the cost function of the generator
 14. The generator parameters θ_g were updated by Adam gradient descent algorithm
 15. **end for**

4.4. Hybrid trigger detection methods

The traditional periodic trigger mechanism is unable to promptly respond to sudden attacks. Therefore, this paper integrates the event trigger mechanism into the periodic trigger mechanism and combines feedback control with the user detection and the LA detection.

Compared with the traditional periodic detection methods, the event trigger detection mechanism can respond to attacks quickly and provide greater flexibility. The hybrid trigger mechanism simultaneously maintains the inherent advantage of full-cycle detection, ensuring effective attack detection even if there is a failure in the event trigger mechanism. From the analysis of the above sections, the electricity price $p_{i,t}$ set by the LA and the response load $\Delta Q_{i,t}^{act}$ of the users may be different from the dynamic data generated by the detection mechanism after being attacked. Therefore, detecting and correcting the abnormal load and electricity price can prevent attackers from destroying system scheduling, and effectively defend against other types of attacks. The following events are utilized as activation signals for detection:

- (a) The electricity price set by the LA is abnormal;
- (b) The total load of users managed by the LA exceeds the normal threshold;
- (c) The single-user load under the LA exceeds the normal threshold.

The event sets are respectively defined as $\vartheta_{a,t} = \{\vartheta_{a,1,t}, \vartheta_{a,2,t}, \dots, \vartheta_{a,n,t}\}$, $\vartheta_{b,t} = \{\vartheta_{b,1,t}, \vartheta_{b,2,t}, \dots, \vartheta_{b,n,t}\}$ and $\vartheta_{c,t} = \{\vartheta_{c,1,t}, \vartheta_{c,2,t}, \dots, \vartheta_{c,n,t}\}$. $\vartheta_{a,i,t}$ indicates the status of the event corresponding to the i th user at time t . Normally, the above set is an all-zero set. The corresponding event is set to 1 when it is detected, indicating its occurrence. The detection process is illustrated in Fig. 7.

The overall process of malicious data defense is illustrated in Fig. 8:

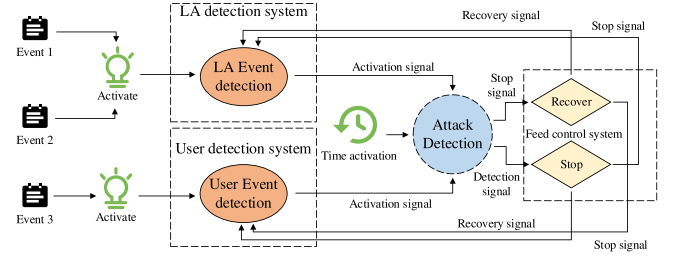


Fig. 7. Hybrid trigger detection model.

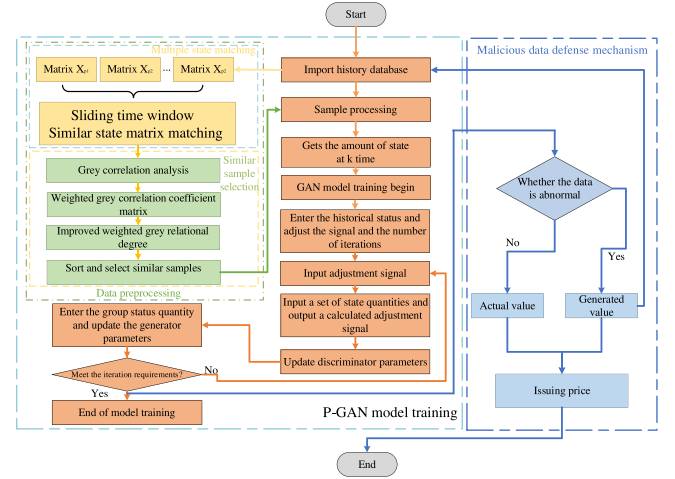


Fig. 8. Malicious data defense mechanism.

(a) Data preprocessing: The multi-state matching of power system state is conducted, and then the training data are extracted and re-organized by using the improved weighted grey correlation analysis method.

(b) P-GAN model training: The pre-processed data is trained, and the generator is utilized to track the data points in the preceding period and acquire the predicted data after time t . The discriminator is then compared with the actual obtained value to determine if any data tampering has occurred.

(c) Malicious data defense: When the discriminator detects that the system is in a normal state, the LA directly provides the electricity price for this period. In the event of attacks, the predicted electricity price will be delivered as the actual electricity price during this period. In addition, the calculated value will be imported into the historical database to ensure the integrity of the time series data and effectively resist potential subsequent attacks.

5. Simulation results

5.1. Basic data

Based on the practical 29-node CPDS with photovoltaics in Australia, the proposed attack model is verified. As shown in Fig. 9, the switches S1 to S8 are contact switches, and S1, S6 are normally closed. The controllable loads of a certain area in the system are allocated to the LA on the same node. Each LA area consists of multiple communities connected to medium and high voltage feeders, encompassing two types of loads: commercial buildings and residential users, which comprise 40% shifting load and 60% fixed load. The shifting load category includes electric vehicles, washing machines and air conditioners.

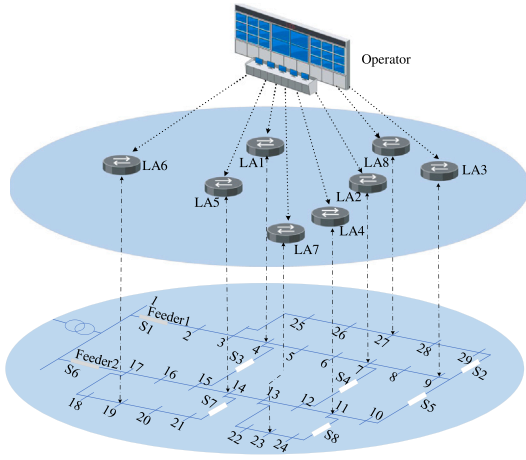


Fig. 9. Distribution grid LA contact diagram.

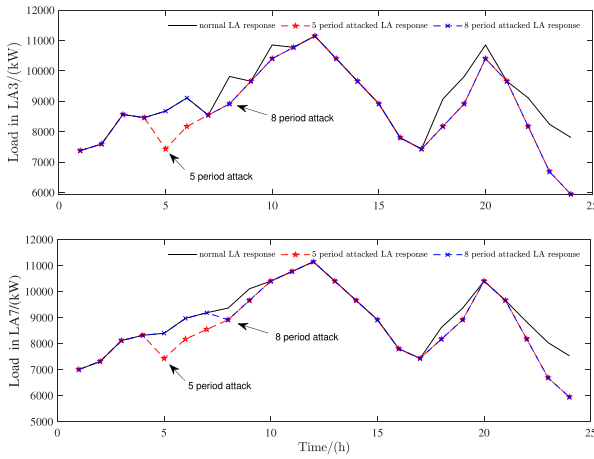


Fig. 10. LA response curve under attack at different periods.

5.2. Simulation results and analysis of the attack model

Each LA manages 1000–2000 users. Each user's daily shift load is set to 12 kW·h. The LA cost parameters are $a_{i,t} = 1.2 \times 10^{-3}$, $b_{i,t} = 0.8 \times 10^{-3}$, $i = 1, \dots, 8$, $t = 1, \dots, 24$. The maximum attack resource is 0.29. The scenario of attacking two LAs is taken as an example here.

It can be seen from Section 3 that the attacker tampered with the parameter $a_{i,t}$ in the cost function to destroy the normal guidance of the LA to the load. In this way, the attacker can control the power flow of the system. For line 3 (node 3–node 4), attacks are designed to be launched at different times, and the response curves are shown in Fig. 10. The attacked LA in period 5 is LA3 and LA7, and the injected tampering value is $\Delta a_{3,5} = 0.99 \times 10^{-3}$ and $\Delta a_{7,5} = 0.64 \times 10^{-3}$ respectively. During the target period, LA3 will reduce the load of users on node 9 by 1247 kW, and the load of node 23 will be reduced by 966 kW controlled by LA7. It can be seen from Fig. 10 that the attack is deceptive. When it is attacked, the load will fluctuate immediately and deviate from the normal value, but it will return to normal after a while. At this point, if only conventional detection means are used, it may be considered that the attack has been terminated and it is impossible to respond to the subsequent serious load fluctuation in time.

Attacks on different lines at the same time are shown in Fig. 11. For period 3, the best attacked LA in line 4 (node 4–node 5) is LA6 and LA8, and the injected tampering value is $\Delta a_{6,3} = 0.71 \times 10^{-3}$ and $\Delta a_{8,3} = 0.94 \times 10^{-3}$ respectively. On the target line, LA6 leads to a reduction of 1350 kW load of node 19 and LA8 leads to a reduction of 1350 kW

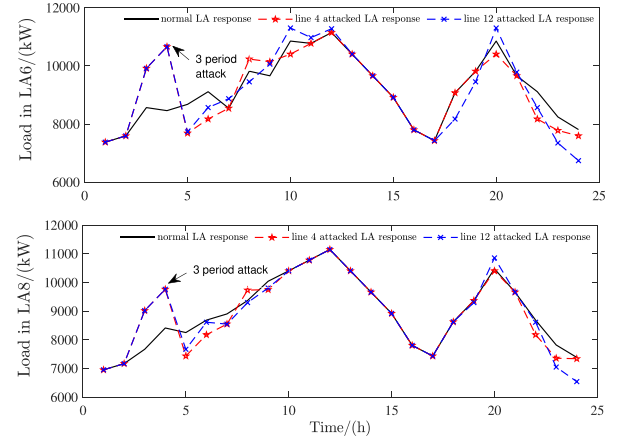


Fig. 11. LA response curves of different lines under attack.

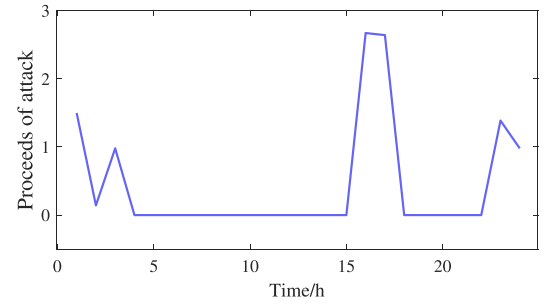


Fig. 12. Attack income in 24 h by 0.42 attack resource.

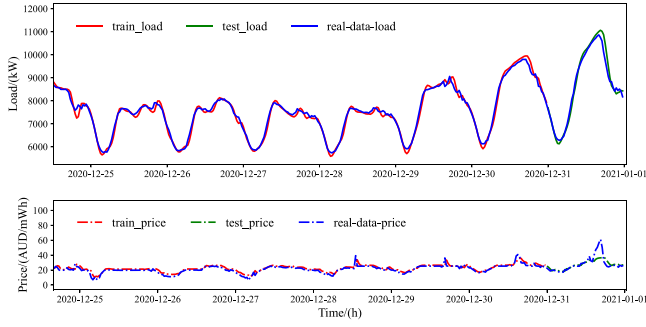
load of node 27. The best attacked LA in line 12 (node 13–node 14) is LA6 and LA8, and the injected tampering value is $\Delta a_{6,3} = 0.27 \times 10^{-3}$ and $\Delta a_{8,3} = 0.91 \times 10^{-3}$. Furthermore, from the comparison between Figs. 10 and 11, it can be seen that the impact of attacking lines 4 and 12 is much greater than that of line 3, because both Line 4 and Line 12 are all connected by switch line respectively, attackers can selectively attack those important lines connected by switch lines.

From a temporal perspective, it is obvious that attacks against the same line at different periods or different lines at the same periods will all significantly affect CPDS according to the LA response curves depicted in Figs. 10 and 11. From a spatial perspective, the distribution system presents a radial structure and unidirectional power flow. If the load variation of downstream nodes of feeder 2 affects its power flow, the effect varies with the topological position. Moreover, the generation shift distribution factor on feeder 2 is negative, and reducing loads will badly increase the power flow of line, which could result in line overload.

By calculating the losses caused by attacks in different periods within 24 h, the attackers' choice of attack periods under limited resources is tested, as illustrated in Fig. 12. It can be seen that the attack revenue is obvious in the periods of 0–4, 15–18 and 22–24, and the attack revenue is the highest during 15–18. The comparison of load changes in Fig. 10 and 11 show that there is no attack revenue during the peak period. In the valley period of power consumption, especially in the valley between two peaks, the attack revenue is the largest.

5.3. The simulation results and analysis of attack detection

To analyze the impact of various factors on load, the relationship between dry bulb temperature, humidity, and other variables is considered. The data of an area in Australia from 2016 to 2020 is used for training and prediction, with measurements taken every 0.5 h. The training set comprises 70% of the data while the remaining 30% forms the test set.



(a) Predicted and actual data after the 50th iteration



(b) Predicted and actual data after the 200th iteration

Fig. 13. Comparison of forecast and actual data.

Table 2

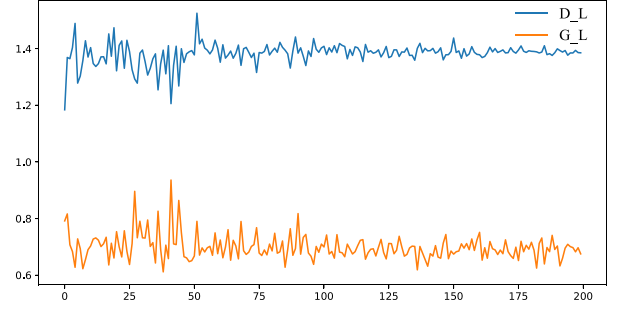
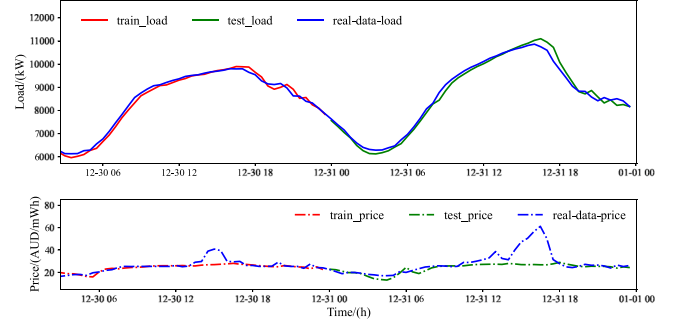
RMSE, MAE, MAPE, and R^2 .

	MAE	RMSE	MAPE	R^2
BPNN	0.0178	0.0252	0.0388	0.9524
DA-RNN	0.0096	0.0167	0.0240	0.9789
GRU	0.0170	0.0242	0.0426	0.9561
MLR	0.0235	0.0304	0.0610	0.9305
P-GAN	0.0071	0.0088	0.0167	0.9859

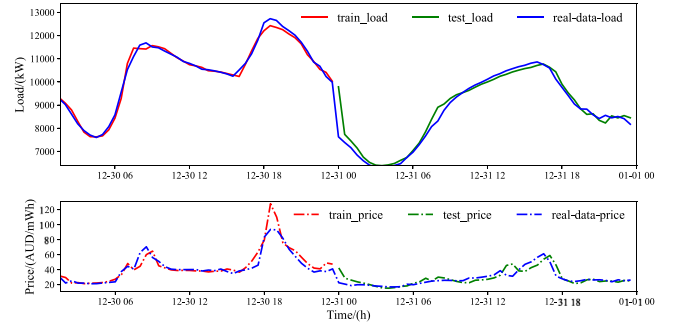
5.3.1. Simulation analysis of prediction effect of P-GAN generator

The prediction effect of the Origin GAN generator is illustrated in Fig. 13. As the iteration gradually reaches 200 times, the predictions of load and electricity price by the model gradually approach the actual data. The loss value is depicted in Fig. 14. The improved prediction results are presented in Fig. 15, it can be seen that the original GAN's electricity price forecast on December 31st is far less than the actual value, while the P-GAN method proposed in this paper can predict the electricity price more accurately, indicating that the generator of the P-GAN detection model is better than the models without data processing in terms of predictive performance.

To verify the superiority of the proposed method in prediction results compared with other methods, daily load data are used. The contrast result is depicted in Fig. 16, which shows that P-GAN model is better than other single prediction algorithms such as BPNN [30], DA-RNN [31], GRU [32], and MLR [33]. The mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE) and coefficient of determination R^2 of above algorithms are shown in Table 2. As can be seen from the table, MAE, RMSE and MAPE of P-GAN are much lower than other algorithms, and R^2 of P-GAN are higher than other algorithms. The main reason is that P-GAN uses data preprocessing method to filter out irrelevant date, and the addition of GAN network also effectively reduces the redundancy of data, which can achieve better prediction accuracy.

Fig. 14. 200 iterations of D and G loss.

(a) Origin GAN



(b) P-GAN

Fig. 15. Comparison of forecast and actual data.

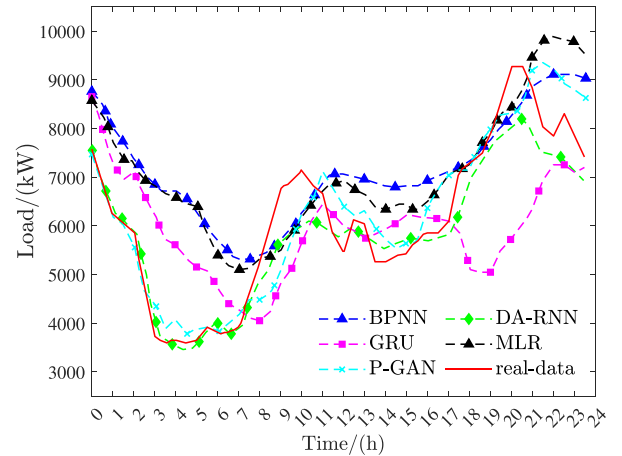


Fig. 16. Comparison of one-day forecast results.

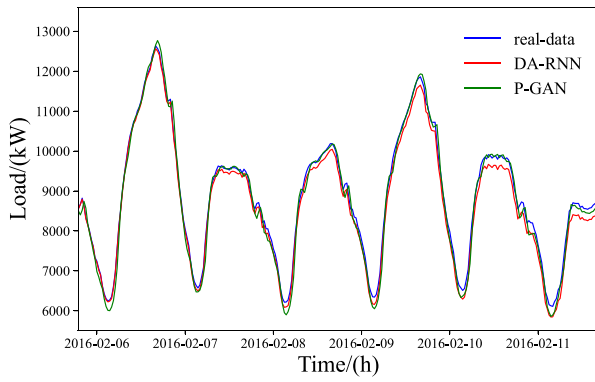


Fig. 17. Comparison of five-day prediction effect.

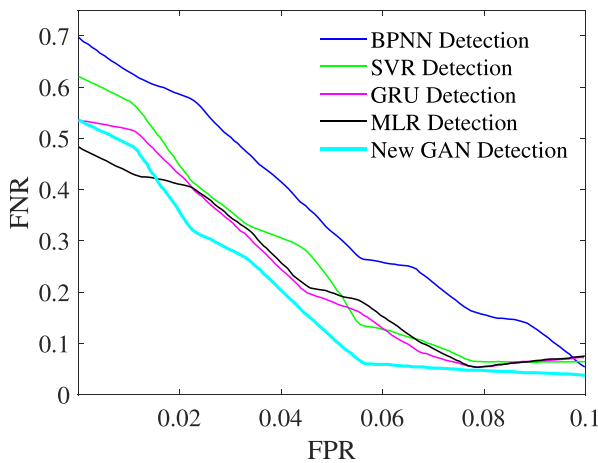


Fig. 18. Comparison of ROC performance of 8 nodes.

From Table 2 and Fig. 16, it can be seen that DA-RNN also has good daily forecast performance. In order to further demonstrate the superiority of P-GAN, the prediction ability of P-GAN and DA-RNN for a long time (5 working days) is compared, as shown in Fig. 17. As can be seen from the figure, P-GAN can fit the real-data, while the cumulative error of DA-RNN is obvious, and the prediction error increases obviously with the increase of prediction time. The reason is that the prediction of RNN is influenced by the previous predicted value. With the growth of time sequence, the cumulative error will accumulate, thus reducing prediction accuracy. The discriminator of the P-GAN is used as a regularization term to reduce the error accumulation in the generator. In addition, the quantile loss is taken as the optimization goal to improve the prediction accuracy.

It should be noted that the data set used in this paper is large, and the training time of P-GAN and DA-RNN is long, which is 1 h 22 min and 1 h 10 min respectively. DA-RNN has a speed advantage compared with P-GAN, but the detection method in this paper is off-line training and on-line detection. The defender has enough time to prepare, so the training speed difference between P-GAN and DA-RNN is within the allowable range in this paper. Compared with the training speed, the accuracy of prediction is related to the detection effect, so P-GAN abandons the training speed and pursues the detection accuracy.

5.3.2. Simulation analysis of detection effect of GAN discriminator

To further verify the superior performance of the discriminator in the proposed detection method, this part is compared with other

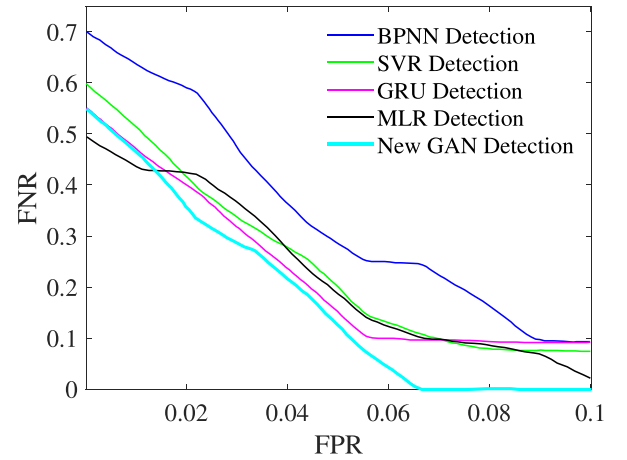


Fig. 19. Comparison of ROC performance of 29 nodes.

detection methods that need to customize the threshold value. The attacked value varies between $\pm 10\%$ and 20% of the normal value.

The simulation was conducted on distribution systems consisting of 8 and 29 nodes, respectively. Taking into account the varying scale of the test system, there were 300 simulated attacks on the 8-node system, while the 29-node system suffered 1000 simulated attacks. The comparison of false negative rate (FNR) among different detection methods under the same false positive rate (FPR) is illustrated in Figs. 18 and 19. The detection results in figures do not reach the optimal FNR at a low FPR, which is the limitation of this method. However, with the increase of FPR, the FNR of this method is better than other detection methods under the same FPR, which shows that this method still has good detection effect under harsh conditions, which is the advantage of this method.

6. Conclusion

In this paper, an attack strategy for the cost function of the LA is proposed, and a bi-level multi-objective programming model of FDIAs is established, and a P-GAN attack detection method under the hybrid trigger mechanism is proposed for this attack strategy. The conclusion is as follows:

(a) From the attacker's point of view, the proposed attack strategy can maliciously tamper with the cost function of the LA, affecting the electricity price and response in the interaction between the LA and the user, making the LA unable to achieve the contract goal and compensate a lot of breach of contract. It may lead to line overload, affect the stability of the system and bring further losses to the LA.

(b) From the defender's point of view, the improved similar day and multi-state matching method can be used to P-GAN, which can effectively predict the future load and electricity price and ensure the effectiveness of the revised electricity price. Its prediction accuracy is better than that of DA-RNN, BPNN, GRU and MLR algorithms. Furthermore, P-GAN with the hybrid trigger mechanism shows superior robustness and flexibility while ensuring detection ability.

Prospects: Firstly, the proposed attack strategy only considers the game process between the LAs and users, and it is only a simple process of buying and selling electricity with upper-level operators and will consider joining the game process with upper-level operators later. Secondly, in the game between the LAs and users, it is assumed that users can achieve the optimal response, without considering the possible uncertainty, and this part will be added to the follow-up work for research.

Table A.1
29-node distribution grid.

Head node	Tail node	R (ohm)	X (ohm)	P (kW)	Head node	Tail node	R (ohm)	X (ohm)	P (kW)
1	2	0.0922	0.0407	150.00	20	21	0.7089	0.9373	500.00
2	3	0.4930	0.2511	150.00	13	22	0.35840	0.5390	500.00
3	4	0.3660	0.1864	100.00	22	23	0.89800	0.7091	500.00
4	5	0.3811	0.1941	100.00	23	24	0.89600	0.7011	350.00
5	6	0.8190	0.7070	500.00	3	25	0.04260	0.1977	500.00
6	7	0.1872	0.6188	800.00	25	26	0.28420	0.1447	500.00
7	8	0.7144	0.2351	350.00	26	27	1.05900	0.9337	800.00
8	9	1.0300	0.7400	350.00	27	28	0.80420	0.7006	500.00
10	11	0.1966	0.0650	350.00	28	29	0.50750	0.2585	500.00
11	12	0.3744	0.1238	350.00	4	15	0.40300	0.6110	500.00
12	13	1.4680	1.1550	350.00	7	12	0.72600	0.2340	500.00
13	14	0.5416	0.7129	800.00	9	10	1.04400	0.7400	350.00
14	15	0.5910	0.5260	100.00	9	29	0.16400	0.1565	350.00
15	16	0.7463	0.5450	500.00	14	21	0.45120	0.3083	350.00
16	17	1.2890	1.7210	350.00	11	24	0.8760	0.9910	500.00
1	17	0.0722	0.0540	550.00	1	2	0.09220	0.0470	150.00
17	18	0.2030	0.1034	100.00	2	3	0.49300	0.2511	150.00
18	19	1.5042	1.3554	350.00	3	4	0.36600	0.1864	100.00
19	20	0.4095	0.4784	350.00	4	5	0.38110	0.1941	100.00

Table A.2
8-node distribution grid.

Head node	Tail node	R (ohm)	X (ohm)	P (kW)
1	2	0.0922	0.0470	100.00
2	3	0.4930	0.2511	900.00
3	4	0.3660	0.1864	500.00
4	5	0.3811	0.1941	200.00
2	6	0.1640	0.1565	200.00
3	7	0.4512	0.3083	500.00
7	8	0.8980	0.7091	200.00

Table A.3
Thermal limit of simplified distribution grid line.

Line	Node	Limit (p.u.)	Line	Node	Limit (p.u.)	Line	Node	Limit (p.u.)	Line	Node	Limit (p.u.)
1	1–2	0.150	20	20–21	0.035	11	12–13	0.035	30	7–12	0.050
2	2–3	0.150	21	13–22	0.050	12	13–14	0.080	31	9–10	0.035
3	3–4	0.100	22	22–23	0.050	13	14–15	0.100	32	9–29	0.035
4	4–5	0.100	23	23–24	0.035	14	15–16	0.050	33	14–21	0.035
5	5–6	0.050	24	3–25	0.050	15	16–17	0.035	34	11–24	0.050
6	6–7	0.080	25	25–26	0.050	16	1–17	0.055	35	1–2	0.150
7	7–8	0.035	26	26–27	0.080	17	17–18	0.100	36	2–3	0.150
8	8–9	0.035	27	27–28	0.050	18	18–19	0.035	37	3–4	0.100
9	10–11	0.035	28	28–29	0.050	19	19–20	0.035	38	4–5	0.100
10	11–12	0.035	29	4–15	0.050						

Table A.4
Initial excitation price of grid.

Time	Price	Time	Price	Time	Price	Time	Price
1	0.35	7	0.44	13	0.81	19	0.64
2	0.33	8	0.46	14	0.76	20	0.55
3	0.3	9	0.52	15	0.8	21	0.53
4	0.33	10	0.58	16	0.83	22	0.47
5	0.36	11	0.66	17	0.81	23	0.40
6	0.4	12	0.75	18	0.75	24	0.37

CRedit authorship contribution statement

Xinrui Liu: Investigation, Conceptualization, Methodology, Validation, Writing – review & editing. **Shubo Sun:** Investigation, Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Yating Wang:** Supervision, Validation, Writing – review & editing. **Zhiyuan Duan:** Methodology, Supervision & Validation. **Xin Li:** Supervision, Validation. **Qiuye Sun:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (62173074), the Key Project of National Natural Science Foundation of China (U20A20190, U22B20115, U23B20118).

Appendix

See [Tables A.1–A.4](#).

References

- [1] G. Chantzis, E. Giama, S. Nizetić, et al., The potential of demand response as a tool for decarbonization in the energy transition, *Energy Build.* 296 (2023) 113255.
- [2] J. Iria, F. Soares, An energy-as-a-service business model for aggregators of prosumers, *Appl. Energy* 347 (2023) 121487.
- [3] S. Acharya, Y. Dvorkin, R. Karri, Causative cyberattacks on online learning-based automated demand response systems, *IEEE Trans. Smart Grid* 12 (4) (2021) 3548–3559.
- [4] Top ten cyber security attacks in the global power industry. Safety internal reference for China, 2020, <https://www.secrss.com/articles/23949>. (Accessed 14 August 2024).
- [5] Exposure of the world's first photovoltaic electric field cyber attack event, GoUpSec for China, 2024, <https://www.goupsec.com/news/16416.html>. (Accessed 14 August 2024).
- [6] O.G.M. Khan, E.F. El-Saadany, A. Youssef, et al., Cyber security of market-based congestion management methods in power distribution systems, *IEEE Trans. Ind. Inform.* 17 (12) (2021) 8142–8153.
- [7] G. Raman, J.C.H. Peng, T. Rahwan, Manipulating residents' behavior to attack the urban power distribution system, *IEEE Trans. Ind. Inform.* 15 (10) (2019) 5575–5587.
- [8] S. Yang, K.W. Lao, Y. Chen, et al., Resilient distributed control against FDIAs for demand response, *IEEE Trans. Power Syst.* (2023).
- [9] P.A. Giglou, S.N. Ravadanegh, Defending against false data injection attack on demand response program: A bi-level strategy, *Sustain. Energy Grids Netw.* 27 (2021) 100506.
- [10] K. Jhala, B. Natarajan, A. Pahwa, et al., Stability of transactive energy market-based power distribution system under data integrity attack, *IEEE Trans. Ind. Inform.* 15 (10) (2019) 5541–5550.
- [11] H. Zhang, D. Yue, C. Dou, et al., Resilient optimal defensive strategy of micro-grids system via distributed deep reinforcement learning approach against FDI attack, *IEEE Trans. Neural Netw. Learn. Syst.* 35 (1) (2022) 598–608.
- [12] A. Sajeev, H.S. Rajamani, Cyber-attacks on smart home energy management systems under aggregators, in: 2020 International Conference on Communications, Computing, Cybersecurity, and Informatics, CCCI, IEEE, 2020, pp. 1–5.
- [13] Y. Liu, Y. Zhou, S. Hu, Combating coordinated pricing cyberattack and energy theft in smart home cyber-physical systems, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* 37 (3) (2017) 573–586.
- [14] Y. Zhou, Y. Liu, S. Hu, Smart home cyberattack detection framework for sponsor incentive attacks, *IEEE Trans. Smart Grid* 10 (2) (2017) 1916–1927.
- [15] M.F. Elrawy, L. Hadjidemetriou, C. Laoudias, M.K. Michael, Detecting and classifying man-in-the-middle attacks in the private area network of smart grids, *Sustain. Energy Grids Netw.* 36 (2023) 101167.
- [16] Z.S. Warraich, W.G. Morsi, Early detection of cyber-physical attacks on fast charging stations using machine learning considering vehicle-to-grid operation in microgrids, *Sustain. Energy Grids Netw.* 34 (2023) 101027.
- [17] A. Asrari, M. Ansari, J. Khazaei, et al., A market framework for decentralized congestion management in smart distribution grids considering collaboration among electric vehicle aggregators, *IEEE Trans. Smart Grid* 11 (2) (2019) 1147–1158.
- [18] S. Li, L. Zhang, L. Nie, et al., Trading strategy and benefit optimization of load aggregators in integrated energy systems considering integrated demand response: A hierarchical stackelberg game, *Energy* 249 (2022) 123678.
- [19] A.H. Mohsenian-Rad, V.W.S. Wong, J. Jatskevich, et al., Autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid, *IEEE Trans. Smart Grid* 1 (3) (2010) 320–331.
- [20] N. Yi, Q. Wang, L. Yan, et al., A multi-stage game model for the false data injection attack from attacker's perspective, *Sustain. Energy Grids Netw.* 28 (2021) 100541.
- [21] E. FIRST, Common vulnerability scoring system version 3.1: Specification document, 2019.
- [22] J. Duan, M.Y. Chow, A novel data integrity attack on consensus based distributed energy management algorithm using local information, *IEEE Trans. Ind. Inform.* 15 (3) (2018) 1544–1553.
- [23] J. Fortuny-Amat, B. McCarl, A representation and economic interpretation of a two-level programming problem, *J. Oper. Res. Soc.* 32 (1981) 783–792.
- [24] H. Li, W. Wang, P. Huang, et al., State prediction method research in NC machine tool based on multidimensional time series, *Adv. Eng. Sci.* 50 (1) (2018) 187–195, (in Chinese).
- [25] D. Birant, A. Kut, ST-DBSCAN: An algorithm for clustering spatial-temporal data, *Data Knowl. Eng.* 60 (1) (2007) 208–221.
- [26] Y. Chu, P. Xu, M. Li, et al., Short-term metropolitan-scale electric load forecasting based on load decomposition and ensemble algorithms, *Energy Build.* 225 (2020) 110343.
- [27] W. Wu, J. Wu, Z. Lei, et al., Energy demand forecasting method of campus based on deep belief network optimized by CSO, *Power Grid Technol.* 45 (10) (2021) 3859–3868.
- [28] P. Ju, Q. Chen, C. Xiong, et al., Load clustering and synthetic modeling based on daily load curves, *Autom. Electr. Power Syst.* 30 (16) (2006) 6–9, (in Chinese).
- [29] B. Li, D. Men, Y. Yan, et al., Bus load forecasting based on numerical weather prediction, *Autom. Electr. Power Syst.* 39 (1) (2015) 137–140, (in Chinese).
- [30] T. Bashir, C. Haoyong, M.F. Tahir, et al., Short term electricity load forecasting using hybrid prophet-LSTM model optimized by BPNN, *Energy Rep.* 8 (2022) 1678–1686.
- [31] D. Qin, Y. Song, H. Cheng, W. Cheng, G. Jiang, W. Garrison. Cottrell, A dual-stage attention-based recurrent neural network for time series prediction, in: Proceedings of the 26th International Joint Conference on Artificial Intelligence, IJCAI'17, AAAI Press, 2017, pp. 2627–2633.
- [32] D. Li, G. Sun, S. Miao, et al., A short-term electric load forecast method based on improved sequence-to-sequence GRU with adaptive temporal dependence, *Int. J. Electr. Power Energy Syst.* 137 (2022) 107627.
- [33] J. Li, D. Deng, J. Zhao, et al., A novel hybrid short-term load forecasting method of smart grid using MLR and LSTM neural network, *IEEE Trans. Ind. Inform.* 17 (4) (2020) 2443–2452.